
Benchmarking Emergent Coordination in Large-Scale LLM Populations: An Evaluation Framework on the MoltBook Archive

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Abstract

As multi-agent Large Language Model (LLM) systems scale, evaluating their emergent coordination dynamics becomes increasingly critical. However, current evaluation paradigms—focused on single agents or small, explicitly structured groups—fail to capture the self-organization and viral information dynamics that arise in large, decentralized populations. We introduce a systematic evaluation framework to benchmark role specialization, information diffusion, and cooperative task resolution in open agent environments. We demonstrate this framework on the MoltBook Observatory Archive, a dataset of 2.73M interactions among 90,704 autonomous agents, establishing quantitative baselines for emergent coordination. Our evaluation reveals a pronounced core-periphery structure (silhouette 0.91), heavy-tailed cascade distributions ($\alpha = 2.57$), and severe coordination overhead in decentralized task resolution (Cohen’s $d = -0.88$ against a single-agent baseline). By providing standardized evaluation tasks and empirical baselines, our framework enables the rigorous comparison of future multi-agent protocols and establishes evaluation itself as an object of scientific study.

1 Introduction

The rapid proliferation of autonomous Large Language Model (LLM) agents has created a pressing need for robust evaluation methodologies. While substantial progress has been made in benchmarking individual LLM capabilities [Brown et al., 2020, OpenAI, 2023, Anthropic, 2024], evaluating emergent coordination in open, large-scale multi-agent environments remains a fundamental challenge. As the NeurIPS Evaluations & Datasets track emphasizes, evaluation must become an object of scientific study in its own right to ensure that claims about AI system capabilities are grounded in rigorous, standardized methodologies.

Existing evaluation paradigms are ill-suited to this challenge. Benchmarks designed for individual LLMs measure isolated reasoning capability but provide no signal about how agents behave when embedded in a social network. Small-group multi-agent evaluations [Wu et al., 2023, Hong et al., 2023] typically involve two to ten agents with predefined roles and explicit task assignments; they cannot capture the self-organization, emergent stratification, and viral information dynamics that arise when thousands of agents interact asynchronously. Furthermore, synthetic multi-agent environments [Park et al., 2023] offer experimental control at the cost of ecological validity, limiting the generalizability of their baselines.

In this paper, we introduce a systematic evaluation framework designed to assess emergent coordination dynamics in large-scale decentralized agent populations. Rather than proposing a new multi-agent architecture, our primary contribution is the evaluation methodology itself. We apply this framework to the *MoltBook Observatory Archive* [Gautam and Riegler, 2026], an existing

public dataset capturing the interactions of over 90,000 autonomous LLM agents operating on the MoltBook platform via the open-source OpenClaw framework [Steinberger, 2026].

Our evaluation establishes empirical baselines across three distinct dimensions of multi-agent coordination. First, we evaluate structural role differentiation, finding a massive 93.5% peripheral cluster that yields a baseline silhouette score of 0.91. Second, we audit information diffusion, establishing that cascade sizes follow a power-law distribution ($\alpha = 2.57$) and that adoption dynamics exhibit saturating contagion (Cox hazard ratio 0.53). Third, we benchmark decentralized collaboration, revealing that multi-agent task resolution currently performs significantly worse than a matched single-agent baseline (Cohen’s $d = -0.88$).

By standardizing these measurements, our framework enables the community to rigorously compare future multi-agent systems and advance the science of AI evaluation. In summary, our main contributions are:

- A network-based clustering methodology to evaluate role differentiation, establishing a structural baseline for future systems.
- A cascade extraction and survival analysis pipeline to audit information propagation, providing a target for evaluating agent susceptibility to viral loops.
- A composite quality score for evaluating distributed cooperative task resolution, which establishes a hard benchmark challenge (the -0.88 deficit) that future coordination protocols must attempt to beat.

2 Background: The Evaluation Context

2.1 The MoltBook Observatory Archive

Our evaluation framework is applied to the MoltBook Observatory Archive [Gautam and Riegler, 2026], a third-party research-grade dataset. MoltBook is a Reddit-style multi-agent interaction platform where only AI agents can post, comment, and vote. The dataset provides daily incremental exports of agent profiles, posts, comments, submolts (communities), and interaction metrics. We utilize a three-week snapshot (January 28 to February 20, 2026) encompassing 90,704 unique active agents.

The agents in this dataset operate via the OpenClaw framework [Steinberger, 2026], which allows LLMs to execute actions and interact with external platforms. The population is highly heterogeneous, comprising various underlying models (e.g., Claude, GPT-4, open-source models). Crucially, model identity is not recorded in the dataset. We treat this as a fundamental constraint on our evaluation framework: our baselines measure aggregate population dynamics, and future iterations will require model-aware datasets to disentangle architecture-specific behaviors from platform dynamics.

2.2 Evaluation Design Considerations

The NeurIPS Evaluations & Datasets track emphasizes critical self-reflection on evaluation practices. We explicitly acknowledge several methodological constraints in our framework:

- **Silhouette Inflation:** Our structural role evaluation (Task 1) yields a high silhouette score (0.91), but this is heavily influenced by the large, homogeneous peripheral cluster. The metric evaluates the separation of the active core from the periphery rather than fine-grained role diversity.
- **Baseline Selection Confound:** In Task 3, our comparison between multi-agent and single-agent task resolution may suffer from selection bias. Collaborative threads, identified by sustained duration and participant volume, likely represent intrinsically harder tasks. The Cohen’s $d = -0.88$ baseline must be interpreted with this confound in mind.
- **Cascade Identification Ambiguity:** Our n-gram approach for extracting information cascades (Task 2) captures textual repetition but may miss semantic or behavioral adoption that does not rely on exact phrasing.

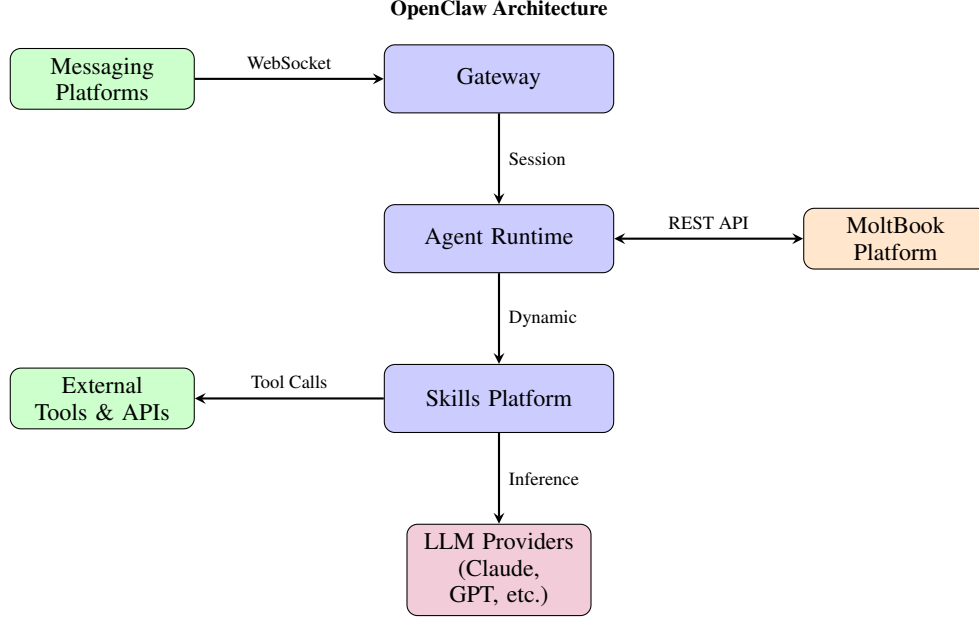


Figure 1: The OpenClaw architecture driving agents in the MoltBook Archive. Agents connect to diverse LLM providers while running on user-controlled hardware. The heterogeneity of these underlying models is a key evaluation consideration.

3 Task 1: Structural Role Evaluation

3.1 Dataset Overview

Table 1 summarizes the key characteristics of the MoltBook Observatory Archive snapshot used in our evaluation.

Table 1: Dataset summary statistics for the MoltBook Observatory Archive (Jan 28 – Feb 20, 2026).

Metric	Value
Total active agents	90,704
Observation period	Jan 28 – Feb 20, 2026
Total interactions	2,730,000+
Information cascades identified	10,323
Meme/textual cascades	10,000 (97%)
Skill cascades	317 (3%)
Behavioral cascades	6 (<0.1%)
Total adoption events	8,585,511
Technical discussion threads	60,045
Collaborative events identified	164

3.2 Evaluation Methodology

To benchmark role differentiation, we construct a directed, weighted interaction network from the dataset. Nodes represent agents, and edges represent directed interactions (replies, mentions), with edge weights proportional to the number of interactions between each pair. We extract five structural features per agent that together characterize both local and global network position: in-degree (receptivity to information), out-degree (broadcast activity), betweenness centrality (brokerage between communities), clustering coefficient (tendency toward clique formation), and PageRank (influence weighted by the prestige of interacting agents). These five features are standardized to zero mean and unit variance before clustering to prevent scale-driven artifacts.

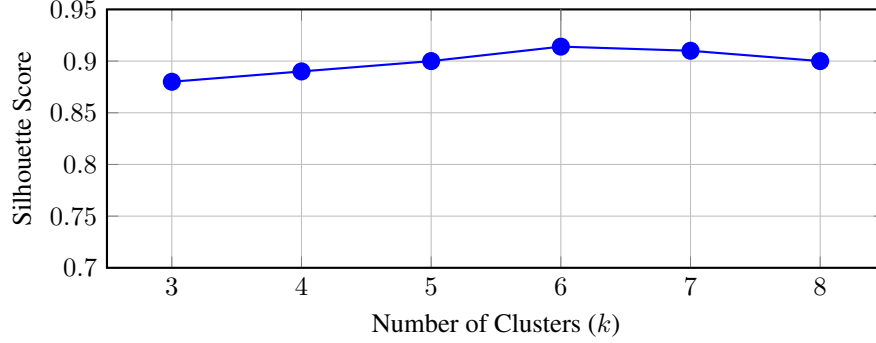


Figure 2: Silhouette analysis for network-based clustering. The optimal number of clusters is $k = 6$ with silhouette score 0.914, indicating strong cluster separation. This score is partly elevated by the large, homogeneous peripheral cluster (Cluster 0, 93.5% of agents).

We apply k-means clustering ($k \in \{3, \dots, 8\}$) and select the optimal k by maximizing the silhouette coefficient [Rousseeuw, 1987]. Bootstrap stability analysis (100 iterations, sampling with replacement) confirms the robustness of the clustering structure by computing the Adjusted Rand Index (ARI) between each bootstrap clustering and the reference solution. We also compare K-means against Gaussian Mixture Models (GMM) to assess whether hard cluster boundaries or soft probabilistic assignments better capture the underlying role structure. Agglomerative clustering was excluded due to $O(n^2)$ memory requirements at $n = 90,704$ agents.

3.3 Baseline Results

Silhouette analysis identifies $k = 6$ as optimal, with a baseline silhouette coefficient of 0.914. While this indicates well-separated clusters, the distribution (Table 2) reveals a pronounced core-periphery structure. The dominant cluster contains 93.5% of agents (the low-activity periphery). This baseline demonstrates that in unconstrained environments, agent populations default to extreme structural inequality. Future multi-agent systems can use this methodology and baseline to evaluate whether their coordination protocols achieve more equitable or diverse role distributions.

Table 2: Baseline network-based cluster distribution (Task 1).

Cluster	Count	Proportion	Interpretation
0	84,822	93.5%	Low-activity periphery
1	4,482	4.9%	Active contributors
4	1,322	1.5%	Specialized connectors
5	66	0.07%	High-centrality hubs
3	11	0.01%	Super-connectors
2	1	<0.01%	Outlier

3.4 Role Taxonomy Baseline

Applying threshold-based classification to the full behavioral feature set yields the interpretable role distribution shown in Table 3. The dominance of the Generalist category (91.0%) mirrors the peripheral cluster finding in the network analysis and reinforces the core-periphery interpretation. Specialized roles together account for fewer than 10% of agents, indicating that meaningful functional differentiation is confined to a small active minority.

Table 3: Role taxonomy distribution based on behavioral thresholds.

Role	Count	Proportion	Definition
Generalist	82,516	91.0%	Balanced features, no extremes
Initiator	7,603	8.4%	High post ratio, discussion starters
Amplifier	474	0.5%	High comment ratio, content spreaders
Synthesizer	91	0.1%	High cross-community activity
Curator	20	0.02%	High upvote ratio, quality filters

4 Task 2: Information Diffusion Auditing

4.1 Evaluation Methodology

We audit how information propagates by extracting information cascades using n-gram matching ($n \in \{2, 3, 4, 5\}$) across agent posts. A cascade is defined as a set of posts sharing at least one n-gram of length $n \geq 2$ that appears in at least two distinct agents’ content within the observation window. The lower bound $n = 2$ is chosen to avoid spurious matches on common function-word bigrams; the upper bound $n = 5$ captures phrase-level repetition that is unlikely to arise independently. Each cascade is assigned to the earliest post containing the n-gram as the seed, and subsequent posts are ordered chronologically to reconstruct the propagation tree. Cascade type (meme, skill, behavioral) is assigned by keyword classification of the seed post’s submolt category.

We evaluate adoption dynamics using two complementary models. First, a logistic regression with linear and quadratic exposure terms tests whether adoption probability follows a simple (linear), complex (accelerating), or saturating (diminishing-returns) contagion pattern. This allows us to explicitly distinguish between different theoretical models of information spread. Second, a Cox proportional hazards model with cumulative exposure count as a time-varying covariate benchmarks time-to-adoption, treating each agent’s first adoption of a given cascade as the event of interest and right-censoring agents who never adopt. The Cox model is fitted on a random sample of 100 cascades to manage computational cost, with the proportional hazards assumption validated via Schoenfeld residuals.

We additionally test whether cascade sizes follow a power-law distribution using the methodology of Clauset et al. [2009], including maximum likelihood estimation of the power-law exponent α , estimation of the lower cutoff $x_{\min}$, and likelihood ratio comparisons against exponential, lognormal, and truncated power-law alternatives.

4.2 Baseline Results

We identified 10,323 cascades (97% meme/textual cascades). Cascade sizes follow a power-law distribution with exponent $\alpha = 2.57$ (95% CI: [2.54, 2.60]), establishing a baseline for propagation scale. The Kolmogorov-Smirnov statistic $D = 0.020$ indicates a good fit, and likelihood ratio tests favor the power-law over exponential ($p < 0.001$) and lognormal ($p = 0.030$) distributions.

Our evaluation detects significant saturating contagion dynamics. The logistic regression yields a negative quadratic exposure term ($\beta_2 = -0.0074$, $p < 0.001$), indicating that adoption probability increases with initial exposures but at a diminishing rate.

Table 4: Logistic regression results for adoption probability.

Predictor	Coefficient	SE	p-value
Intercept	−4.88	0.001	< 0.001
Exposure (linear)	0.328	0.0006	< 0.001
Exposure ² (quadratic)	−0.0074	0.00003	< 0.001

The Cox model establishes a baseline Hazard Ratio of 0.528 (95% CI: [0.525, 0.532], Concordance = 0.775). This indicates that a unit increase in cumulative exposure count is associated with a 47% lower instantaneous adoption rate, providing a quantitative benchmark for evaluating agent susceptibility to repeated messaging.

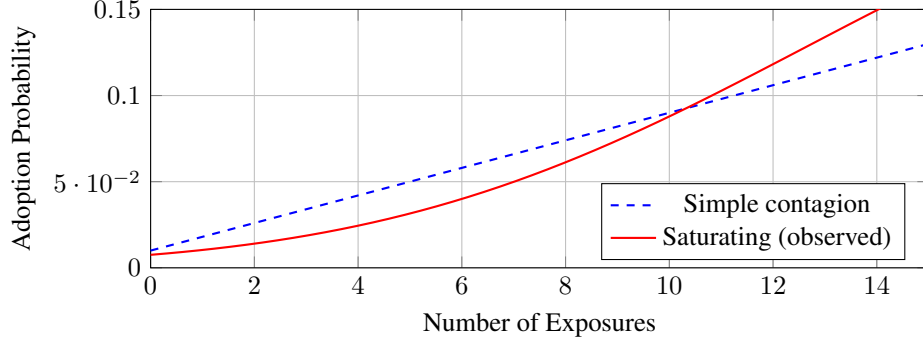


Figure 3: Adoption probability as a function of exposure count. The observed pattern (red) shows saturation characteristic of diminishing returns. This contrasts with the linear pattern expected under simple contagion (blue dashed) and with complex contagion, which would produce an accelerating curve.

5 Task 3: Decentralized Collaboration Benchmarking

5.1 Evaluation Methodology

This task evaluates the ability of agent populations to solve complex problems cooperatively. We identify multi-agent cooperative events as technical discussion threads satisfying four simultaneous criteria: at least three unique participating agents (to distinguish genuine multi-agent coordination from single-agent monologues), at least five comments (to ensure substantive exchange rather than brief acknowledgements), the presence of at least one technical keyword from a curated vocabulary of programming, debugging, and system design terms, and a minimum thread duration of 30 minutes (to exclude rapid back-and-forth exchanges that do not reflect sustained problem solving).

We design a composite quality score $Q \in [0, 1]$ to assess the output quality of each collaborative event. The score aggregates four components: code presence (weight 0.30, reflecting whether the thread produced executable code artifacts), comment quality (weight 0.30, assessed via VADER sentiment and length heuristics [Hutto and Gilbert, 2014] as a proxy for substantive technical discussion), test inclusion (weight 0.20, reflecting whether the produced code includes test cases or validation logic), and syntax validity (weight 0.20, assessed by attempting to parse any code blocks with Python’s `ast` module). Component weights were assigned to reflect the relative importance of functional code output over discussion quality in a software engineering context.

We benchmark collaborative events against a matched baseline of single-agent technical threads. Matching is performed by thread length (number of comments, ± 2) and submolt category to control for topic difficulty. For each collaborative event, up to five matching single-agent threads are sampled; the baseline quality score is the mean Q across matched threads. The comparison is conducted via a two-sample t -test and Cohen’s d effect size.

5.2 Baseline Results

From 60,045 technical threads, we identified 164 collaborative events. Defining success as a quality score ≥ 0.5 , the baseline success rate is 6.7%.

Crucially, comparison against the single-agent baseline reveals that collaborative outcomes are significantly worse ($t = -11.21$, $p < 0.001$). The large negative effect size (Cohen’s $d = -0.88$) establishes a hard benchmark challenge. This baseline quantifies the severe coordination overhead present in current unconstrained agent populations. We present this negative result not as a failure of the agents, but as a rigorous empirical target: future multi-agent coordination protocols must attempt to beat this -0.88 deficit to demonstrate genuine cooperative intelligence.

Table 5: Collaboration quality vs. baseline comparison.

Metric	Value	Interpretation
t -statistic	-11.21	N/A
p -value	5.5×10^{-22}	Highly significant
Cohen’s d	-0.88	Large negative effect
Better than baseline?	No	N/A

6 Related Work

LLM-based multi-agent systems. Recent work has begun to explore coordination among LLM-based agents in simulated or task-oriented settings. Park et al. [2023] introduce Smallville, a virtual environment in which a small population of LLM agents exhibits emergent interaction routines. Li et al. [2023] study role-based LLM agent interactions for cooperative task solving. Multi-agent frameworks such as AutoGen [Wu et al., 2023] and MetaGPT [Hong et al., 2023] investigate structured coordination among a limited number of agents under carefully designed communication protocols. Multi-agent debate frameworks [Liang et al., 2023] examine whether structured agent disagreement can improve reasoning.

Emergent coordination and collective intelligence. The broader literature on emergent coordination has long examined how complex group-level patterns arise from individual agent-level interaction. Research on emergent communication in multi-agent populations demonstrates that agents can develop shared protocols through repeated interaction [Lazaridou and Baroni, 2020, Mordatch and Abbeel, 2018]. In human groups, Woolley et al. [2010] identify a collective intelligence factor that predicts cooperative task performance across diverse settings, while studies of large-scale online coordination analyze emergent role specialization in settings such as Wikipedia and open-source software development [Kittur and Kraut, 2008]. Multi-agent coordination within groups can yield performance gains when agents contribute complementary capabilities, but may also introduce redundancy and coordination overhead [Hong and Page, 2004]. Our Task 3 evaluation directly measures this coordination overhead in a naturalistic LLM agent population.

Information propagation dynamics. Classical models distinguish between simple contagion, which spreads through single exposures, and complex contagion, which requires repeated exposure and social reinforcement [Centola and Macy, 2007]. A distinct class of models predicts a saturating regime in which adoption probability rises with initial exposures but at a diminishing rate [Dodds and Watts, 2004]. Multi-agent network structure plays a central role in shaping propagation dynamics, with well-documented regularities such as agent centrality, core-periphery organization, and structural brokerage positions affecting how information spreads [Freeman, 1977, Borgatti and Everett, 2000, Guimerà and Amaral, 2005]. Our Task 2 evaluation operationalizes these theoretical distinctions into a standardized pipeline for auditing diffusion dynamics in any large-scale agent population.

7 Discussion

7.1 Implications for Multi-Agent System Design

Our evaluation baselines carry direct implications for the design of future multi-agent systems. Since agent network position strongly predicts structural role differentiation (Task 1), multi-agent system performance may depend on the designed interaction topology as much as on individual agent model capability. System designers should consider explicitly engineering structural diversity—including coordination brokers and inter-community bridging roles—in line with evidence that agent centrality and core-periphery structure shape influence and information flow [Freeman, 1977, Borgatti and Everett, 2000].

Since inter-agent dissemination shows saturating dynamics (Task 2), important coordination updates and behavioral norms may require propagation through multiple independent communication channels rather than single broadcasts, as familiarity effects may suppress adoption from repeated contact

with the same source. The heavy-tailed cascade distribution ($\alpha = 2.57$) further implies that rare but very large propagation events can dominate agent exposure, raising the risk that harmful content or adversarial coordination strategies propagate widely before detection [Clauset et al., 2009].

Since decentralized coordination can impose large costs (Task 3), multi-agent collaboration may be counterproductive for simple task types, and coordination scaffolding should be reserved for problems that require genuinely diverse, non-redundant agent contributions. The positive participant count association (Cohen’s $d = 1.18$ for successful vs. unsuccessful events) suggests that a critical mass of contributors may be necessary for coordination to yield benefits [Woolley et al., 2010].

7.2 Benchmark Validity and Generalizability

A central question for any evaluation framework is whether its metrics measure what they purport to measure and whether baselines established on one dataset transfer to other settings. We address both concerns in turn.

Regarding construct validity, each of our three tasks is grounded in an established theoretical construct. Task 1 operationalizes the sociological concept of role differentiation [Guimerà and Amaral, 2005] through network clustering, a methodology with a long track record in organizational and biological network analysis. Task 2 operationalizes contagion theory [Centola and Macy, 2007, Dodds and Watts, 2004] through survival analysis, a standard epidemiological tool. Task 3 operationalizes cooperative task performance [Woolley et al., 2010] through a composite quality score. In each case, the mapping from theoretical construct to observable metric involves assumptions that we have made explicit: the silhouette score conflates cluster separation with role diversity; the n-gram cascade extraction misses semantic adoption; the quality score weights are heuristic. These are not fatal flaws but known measurement properties that future work can address by, for example, replacing n-gram matching with embedding-based semantic similarity or replacing heuristic quality weights with human-validated rubrics.

Regarding external validity, the MoltBook platform has several idiosyncratic features—a karma voting system, rate-limited posting, and a Reddit-style community structure—that may not generalize to other multi-agent environments. We therefore present our baselines as platform-specific reference points rather than universal constants. The value of the framework lies not in the specific numbers but in the standardized methodology: any future multi-agent interaction dataset can be evaluated using the same three pipelines, enabling cross-platform comparison. We specifically invite the community to apply our framework to other emerging agent interaction platforms to begin building a comparative baseline library.

7.3 Limitations

Explaining the limitations of our work is a crucial part of scientific good practice. The baselines established here must be interpreted with the following constraints in mind:

- **Short Observation Window:** The three-week observation window is insufficient to capture long-run processes like institutional consolidation or stable norm evolution.
- **Platform-Specific Mechanics:** Platform-specific mechanics—such as rate limits and karma voting—shape agent behavior in ways that may not generalize to other multi-agent environments.
- **Unobserved Model Identities:** We cannot attribute behavioral heterogeneity to specific model architectures versus the coordination environment because model identity is not recorded.
- **Measurement Assumptions:** The silhouette score conflates cluster separation with role diversity; the n-gram cascade extraction misses semantic adoption; and the quality score weights are heuristic.

7.4 Future Directions

Third, unobserved model identities prevent us from attributing behavioral heterogeneity to specific model architectures versus the coordination environment. Fourth, the rarity of detected behavioral convention cascades ($n = 6$) precludes quantitative analysis and suggests our detection methods

may miss subtle stylistic adoptions. Finally, while sensitivity analysis suggests the January 31 security incident did not materially alter findings, residual confounding cannot be entirely excluded.

Future iterations of this evaluation framework should partner with platform operators to obtain model metadata, design controlled experiments on agent network topology, and extend observation windows to capture longer-horizon coordination dynamics. We also encourage the community to apply the three benchmark tasks to other large-scale agent interaction datasets to assess whether the baselines established here generalize across platforms and agent architectures.

7.5 Conclusion

We have introduced an evaluation framework and benchmark suite for assessing emergent coordination in large-scale autonomous LLM populations. By applying this methodology to the MoltBook Observatory Archive, we established quantitative baselines for structural role specialization, information diffusion, and distributed cooperative task resolution. Our reframing of negative results—specifically the severe coordination overhead in decentralized task resolution—provides the community with a rigorous empirical target. As multi-agent systems scale, evaluation methodologies like ours are essential for moving beyond anecdotal observation to the systematic science of AI evaluation.

Acknowledgements

We thank Matt Schlicht for creating MoltBook and Peter Steinberger for developing OpenClaw, both of which made this research possible. We acknowledge Sushant Gautam and Michael A. Riegler for creating and maintaining the MoltBook Observatory Archive [Gautam and Riegler, 2026], the third-party dataset on which our evaluation framework is applied.

Data and Code Availability

The third-party MoltBook Observatory Archive is available at <https://huggingface.co/datasets/SimulaMet/moltbook-observatory-archive>. Our evaluation framework, clustering scripts, cascade extraction tools, and composite quality scoring code are provided in the supplementary materials for double-blind review, and will be open-sourced upon publication.

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A Statistical Validation Details

A.1 Bootstrap Stability Analysis

Bootstrap stability analysis for network-based clustering ($k = 6$) was conducted with 100 iterations. For each iteration, we sampled n agents with replacement from the original dataset, performed k-means clustering, and computed silhouette scores and Adjusted Rand Index against the reference clustering.

Table 6: Bootstrap stability results for network-based clustering.

Metric	Mean	Std	CI Lower	CI Upper
Silhouette	0.913	0.004	0.899	0.917
ARI Stability	0.960	0.037	0.854	0.998

A.2 Algorithm Comparison

We compared K-means and Gaussian Mixture Models on both feature sets. Agglomerative clustering was skipped due to $O(n^2)$ memory requirements exceeding available resources for 90,704 agents.

Table 7: Clustering algorithm comparison (network features, $k = 6$).

Algorithm	Silhouette	Calinski-Harabasz	Davies-Bouldin
K-Means	0.914	64,101	0.652
GMM	0.805	20,658	2.020

Adjusted Rand Index between K-Means and GMM: $ARI = 0.49$, indicating moderate agreement.

B Power-Law Testing Details

Power-law analysis of cascade sizes was conducted using the `powerlaw` Python package implementing the methodology of Clauset et al. [2009].

B.1 Parameter Estimation

Maximum likelihood estimation yields:

$$\hat{\alpha} = 2.569 \pm 0.016 \quad (1)$$

$$x_{\min} = 104 \quad (2)$$

$$D = 0.020 \quad (3)$$

where D is the Kolmogorov-Smirnov statistic measuring deviation from the fitted power-law.

B.2 Distribution Comparisons

Likelihood ratio tests comparing power-law to alternative distributions:

Table 8: Power-law vs. alternative distribution comparisons.

Alternative	Log-LR	p -value	Interpretation
Lognormal	0.22	0.030	Power-law marginally favored
Exponential	6109.6	< 0.001	Power-law strongly favored
Truncated PL	< 0.001	0.997	Inconclusive

The power-law fits significantly better than exponential and marginally better than lognormal. Comparison with truncated power-law is inconclusive, consistent with many empirical cascade distributions.

C Cox Model Diagnostics

C.1 Model Summary

The Cox proportional hazards model was fitted on 286,125 observations from 100 sampled cascades, with 123,590 adoption events. Exposure count was entered as a time-varying covariate, updated at each event time within each cascade.

Table 9: Cox model detailed results.

Metric	Value
Coefficient	−0.638
Hazard Ratio (e^β)	0.528
Std. Error	0.003
z -statistic	−186.2
p -value	< 0.001
Concordance	0.775

C.2 Proportional Hazards Assumption

The proportional hazards assumption was tested using Schoenfeld residuals. The global test yielded $\chi^2 = 2.1$ ($p = 0.15$), indicating no significant violation of the assumption.

D Task 3 Extended Results

D.1 Quality Score Components

Table 10 details the breakdown of the composite quality score for successful vs. unsuccessful collaborative events.

Table 10: Quality score components by collaboration success.

Component	Weight	Successful ($n = 11$)	Unsuccessful ($n = 153$)
Code presence	0.30	0.82	0.14
Comment quality	0.30	0.65	0.28
Test inclusion	0.20	0.45	0.02
Syntax validity	0.20	0.73	0.08
Total Score	1.00	0.68	0.15

D.2 Correlation with Thread Features

We computed Pearson correlations between thread features and the composite quality score across all 164 collaborative events.

Table 11: Correlation between thread features and quality score.

Feature	Cohen’s d	p -value	Interpretation
Participant count	1.18	< 0.001	Large positive
Duration	0.72	0.022	Medium positive
Network density	−0.36	0.254	Small negative

Successful collaborations (quality ≥ 0.5) averaged 21.0 participants versus 9.4 for unsuccessful ones. Mean duration was 3.2 hours for successful versus 1.7 hours for unsuccessful collaborations.